Software Engineering Lab 4

PES1UG24CS579 | Assignment conversation

Assigned project: SETAPESU26/30_donkeykong

Prepared: 09 October 2026, 09:50:09 IST. 20 actual student and assistant messages.

Scope: the three assignment prompts and their conversation, ending with the final response to the Task 4 multiplier prompt. Later game-launch, recording, and upload discussion is excluded as requested.

Source: actual messages retrieved from this chat, reproduced in order without invented replies or summaries. Original Markdown syntax is retained. System metadata, tool calls/results, private reasoning, and other chat branches are excluded.

1. Student

I’m doing my Software Engineering Lab 4 assignment, My SRN is PES1UG24CS579, and my assigned repository is https://github.com/SETAPESU26/30_donkeykong. Please work with the files that have already been set up locally. The original repository is cloned at /Users/zahoormashahir/Documents/ChatGPT/SE/source/30_donkeykong, and my personal lab repository is at /Users/zahoormashahir/Documents/ChatGPT/SE/source/SE-Labs-PES1UG24CS579. The working game is inside the Lab-4 folder of my personal repository. Use /Users/zahoormashahir/Documents/ChatGPT/SE/work/lab4-venv/bin/python to run it; Python 3.12 is already installed there.
Start by reading Lab-4/README.md and game.py, checking the Git status and remotes, My lab requires a 10-second gameplay video before , another one after all changes, updated code, and the chat history exported as a document or PDF. The repository README additionally asks for the shared chat link. Please preserve these requirements throughout our work.
So i have taken the video already, the barrels in the top layer fall down too often making it hard to complete the game, . Find the ladder-descent probability check inside Barrel.update and correct it so each eligible encounter has approximately a 30% chance of descending. Preserve the existing one-decision-per-ladder behaviour and the rest of the game. Check the decision boundary using controlled random values and, if useful, a repeatable larger sample using the actual Barrel logic. Run the game again to check that the fix has not broken normal movement.
Commit this fix separately with a clear message. Explain the cause, the exact change, and what you checked in straightforward language so I can review it. Leave the three feature functions for my next prompts. Keep all assignment changes inside Lab-4 in my personal repository, preserve my earlier labs, and do not push to SETAPESU26 or create a pull request. Wait for my next prompt after finishing this task.

2. Assistant

I’ll read the lab instructions and game code, check both repositories, then make and verify the barrel fix. I’ll keep the assignment edits inside your personal repository’s Lab-4 folder and commit the fix separately.

3. Assistant

The cause is in `Barrel.update`: `random.random() < 0.7` gives each eligible encounter a 70% descent chance. Changing it to `< 0.3` will give the intended 30% chance, while keeping the existing `skip` set that prevents another decision at the same ladder.

I’m using the debugging skill to check the boundary and movement. Lab-4 is currently untracked, so I’ll first commit the supplied files as a baseline, then commit the fix separately. I’ve also noted your recorded before video and the remaining final video, chat export, and shared-link requirements.

4. Assistant

The checks pass: values below `0.3` descend, while `0.3` and above keep rolling. Repeated frames and revisits do not reroll the decision. A seeded sample of 10,000 encounters through the actual `Barrel.update` logic produced 2,988 descents—29.88%.

The movement checks also passed. I’m checking the visible game now; macOS required display access outside the sandbox to open its window.

5. Assistant

The real game completed an 11-second controlled run successfully. Walking, ladder climbing, movement along the slope, jumping and landing, reset, barrel spawning, and rendering all worked.

I’ll now record the results and submission requirements in the Lab-4 README, then commit the fix. The three feature functions remain untouched.

6. Assistant

Committed as `c76e135` — **Fix Lab 4 barrel ladder descent probability to 30 percent**.

The cause was a `< 0.7` check, giving a 70% descent chance. [game.py](/Users/zahoormashahir/Documents/ChatGPT/SE/source/SE-Labs-PES1UG24CS579/Lab-4/game.py:167) now uses `< 0.3`. The existing one-decision-per-ladder behavior is preserved.

Verified:

- Below `0.3` descends; exactly `0.3` and above continue rolling.
- Seven checks passed, including repeat encounters and movement.
- 10,000 seeded encounters produced **29.88%** descents.
- An 11-second real game run passed walking, climbing, jumping, reset, and rendering checks.

Lab-4 was first committed as baseline `fbd1217`, keeping the fix separate. Earlier labs and all three feature functions are untouched. Nothing was pushed; no PR was created.

The README records your before video and the remaining final video, chat export, and shared-link requirements. Waiting for your next prompt.

7. Student

Continue with Tasks 2 and 3 from the README. Implement theme_color(score) so the background starts navy, becomes dark purple at 500 points, and changes to a warmer dark colour at 1000 points. Keep everything readable and make sure resetting restores the original background. Check it and commit this feature separately. Then implement on_barrel_jumped(player, barrel) to show a floating points label above the barrel for about one second. It should appear only once per successful jump and display the actual points awarded, so it will still be correct when we add the multiplier. Keep the existing callback interface and clear temporary effects when restarting. Test it in the game and make a separate commit.
Keep the code simple and consistent with the existing game. Explain what you changed and what you verified, then wait for my next prompt.

8. Assistant

I’ll implement and verify the score-based background first, then commit it separately. After that I’ll add the floating points label, check that it shows the awarded amount once per jump and clears on restart, and make a second commit.

9. Assistant

The background now stays navy below 500 points, turns dark purple at 500, and becomes a warm dark brown at 1000. I checked both thresholds and the rendered scenes; the game elements remain readable. Restarting after a 1000-point win restored navy.

I’ll commit that feature now. For the label, I’ll store the calculated award on the barrel before calling the existing two-argument callback, so it can show the correct amount when the multiplier is added.

10. Assistant

The floating label passed the game checks: a successful jump showed one `+100` label, and a temporary test multiplier correctly produced `+250`. The label moved upward, disappeared after one second, and cleared when restarting while it was visible.

I also reran the seven barrel and movement checks; all still pass. I'll record these results and make the second feature commit.

11. Assistant

Completed in two separate commits:

- **`7b45a51` — Background colours:** navy below 500 points, dark purple at 500, warm dark brown at 1000. Checked threshold values, readability, and that restarting restores navy.
- **`9d6ef41` — Floating points labels:** one label per cleared barrel, rising for about one second. It displays the exact awarded bonus while preserving `on_barrel_jumped(player, barrel)`. Restarting clears temporary effects.

Real game checks passed for `+100` and a test-only `+250` award, single appearance, upward movement, expiry, and reset. All seven existing barrel/movement checks also passed.

Only Lab-4 changed. Task 4 remains untouched, submission requirements remain recorded in the README, and nothing was pushed. Waiting for your next prompt.

12. Student

Now implement Task 4, score_multiplier(score). Award 1x points below 1000 and 2x at 1000 or above, using the score before the jump bonus is added. Display the multiplier in the HUD and make sure the floating label shows the correct bonus. Reset everything properly when R is pressed, test the threshold, and commit this feature separately.

Prepare the chat-history PDF from the actual conversation and give it to me . If you cannot access the full history, ask me for the export rather than inventing it. Update the Lab-4 README with setup instructions, completed tasks and deliverable locations. Keep only useful submission files and leave everything ready for a final code review before pushing.

13. Assistant

I'll add the 1x/2x multiplier and HUD display, check the threshold and reset behavior in the game, then commit it separately. I'll also try to retrieve the actual conversation for the PDF and update the Lab-4 README with setup steps and deliverable locations.

14. Assistant

The local repository already has Task 4 in its own commit, `07dd1d0`, including the HUD and reset changes. I'll verify that implementation and keep the existing commit.

I can retrieve this chat's actual messages, including the earlier bug fix and feature work. I'm using the PDF skill to refresh the PDF from this conversation; the current file contains messages from the other chat branch.

15. Assistant

All ten checks pass. A jump from 900 to 1000 awards `+100`; the next jump awards `+200`. The HUD shows the multiplier for the next jump, and R restores navy, 1x, three lives, and the initial spawn delay while clearing barrels and labels. I also checked that the HUD and both labels fit the game window.

I'll now generate the conversation PDF and update the README. The after video and shared link will remain marked as pending.

16. Assistant

The PDF snapshot contains 15 actual messages from this chat, including all three assignment prompts. I've also confirmed that the before video is exactly 10 seconds and that the repository contains only Lab-4 changes.

The PDF generation and preview processes have stalled. I'm checking them before delivering the file so I can verify both its contents and layout.

17. Assistant

I've checked all four PDF pages and matched every exported message to the retrieved conversation. I'll include the latest available messages in the final snapshot, commit the PDF and README, and open the game changes for review.

Nothing has been pushed. The after video and shared chat URL are the remaining submission items.

18. Assistant

The PDF is complete and verified, but the documentation commit failed because Git cannot read part of the existing packed history. The Task 4 commit is still present.

I'm checking whether I can restore those Git objects from your personal repository without changing any earlier-lab files. The PDF and README changes are safely staged.

19. Assistant

Git's history is restored from an identical, checksum-verified copy from your personal repository, and its integrity check now passes. Earlier-lab files and the separate feature commits are unchanged.

The README and PDF are committed. I'll make one final PDF refresh to include these repair notes, then leave the code ready for review. Nothing has been pushed.

20. Assistant

Task 4 is complete in separate commit **`07dd1d0`**: 1x below 1000, 2x at 1000+, calculated before adding the bonus. The HUD and floating labels show the correct amounts, and R resets everything. **All ten checks passed.**

Your actual conversation PDF is ready: :codex-file-citation{path="/Users/zahoormashahir/Documents/ChatGPT/SE/source/SE-Labs-PES1UG24CS579/Lab-4/chat/PES1UG24CS579_Lab4_Chat_History.pdf" purpose="output"}. It contains 19 verified messages across four pages, through **12:00:03 IST**.

README setup and deliverable details are committed as **`d433761`**. Git history was safely repaired; earlier labs are unchanged. The working tree is clean, and nothing was pushed. The after video and shared chat link remain pending.

- :codex-followup[Review final code]{prompt="Review all Lab-4 code changes before pushing. Do not push or create a pull request."}
- :codex-followup[Verify after video]{prompt="Help me verify and save the genuine ten-second after gameplay video under Lab-4/videos."}
- :codex-followup[Finalize chat evidence]{prompt="Help me obtain this chat's shared link, refresh the actual conversation PDF, and update the Lab-4 README."}